Laser Spectrometry System Development for Next Generation Lunar Rover Missions

WISTLS Sampling Drill Design and Development

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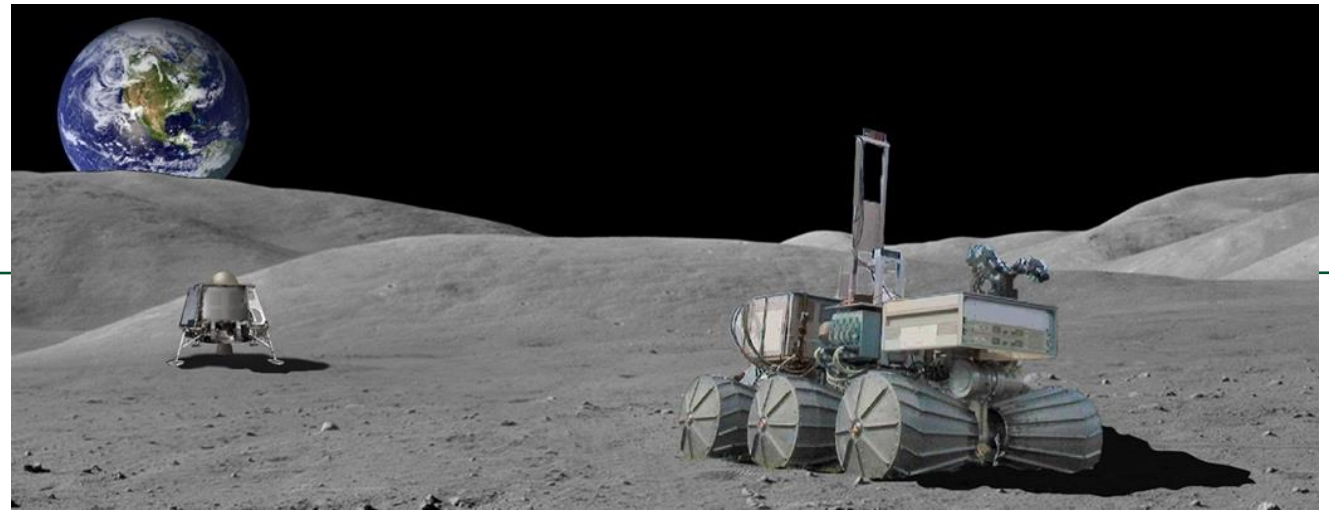
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Kevin Schwarm - JPL

Cal Poly Pomona

WISTLS: Water Isotope Tunable Laser Spectrometer

Exploration & Engineering Benefits



Supports ISRU: Identifies where usable water exists and what purification may be required.

Guides Missions: Informs rover and crew landing sites by locating volatile-rich regions.

Validates Systems: Confirms the WISTLS sampling drill captures representative lunar vapor.

WISTLS: Water Isotope Tunable Laser Spectrometer

Core Objectives

1. **Measure** H_2O, HDO, and D_2O isotopes from heated regolith samples.
2. **Differentiate** solar wind implanted, primordial, and cometary sources.
3. **Quantify** bound vs. free water in lunar regolith.

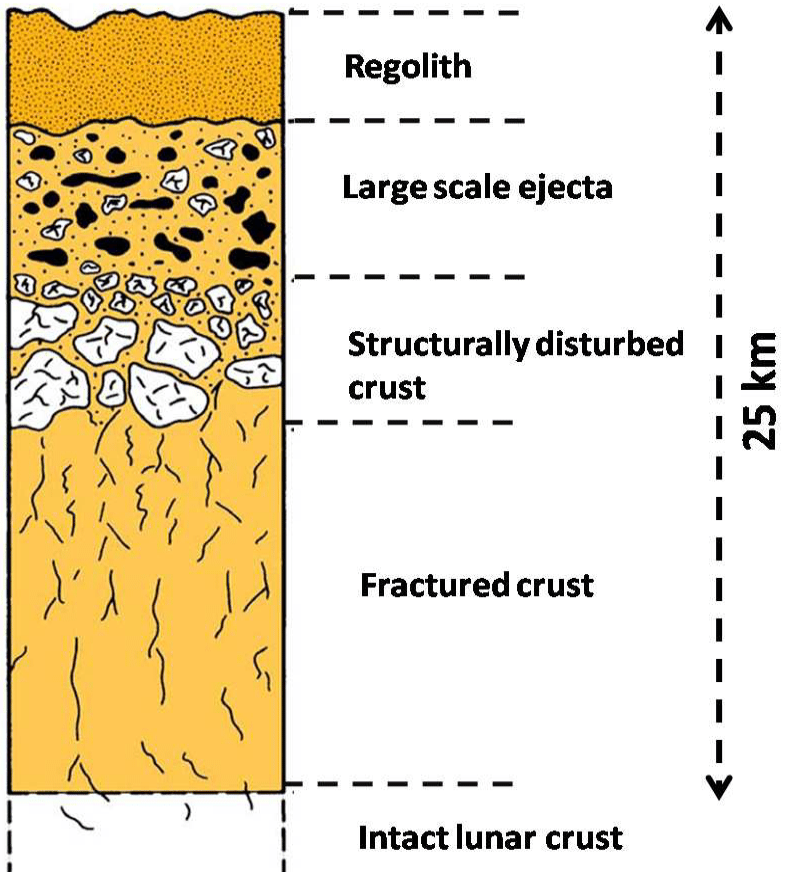
Isotopic Forms of Water

- **H_2O** – Standard water molecule with two hydrogen atoms.
- **HDO** – Semi-heavy water containing one deuterium (heavy hydrogen) atom.
- **D_2O** – Heavy water with both hydrogens replaced by deuterium.

Problem Statement

- NASA mission needs a system to extract and analyze water from lunar regolith and other planetary bodies.
- The system must identify hydrogen and oxygen isotopes
- Key functional needs:

1. Collect regolith from both soft and compact surfaces
2. Vaporize water molecules
3. Transport vapor for analysis



Design Objectives

- Develop a collection system that can vaporize solid water molecules.
- Pulverize regolith substrate
- Keep design modular for future hydrogen isotopic testing.
- A framework for future high-temperature experiments and JPL integration.



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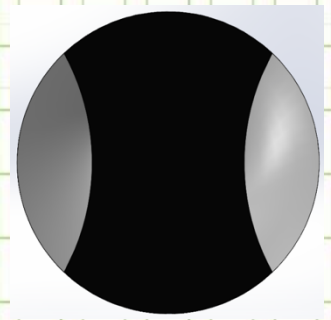
Design and Prototype Process

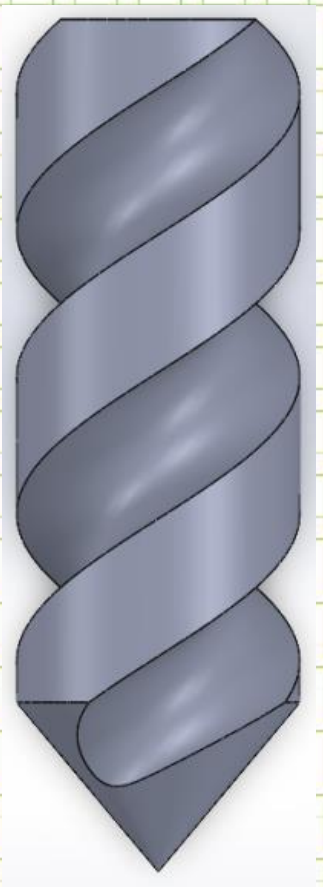
Design 1
Single Helix Auger

- Basic auger concept
- Tested auger's ability to lift sample
- Served as the baseline for future iterations focused on vapor management.

Key Takeaway

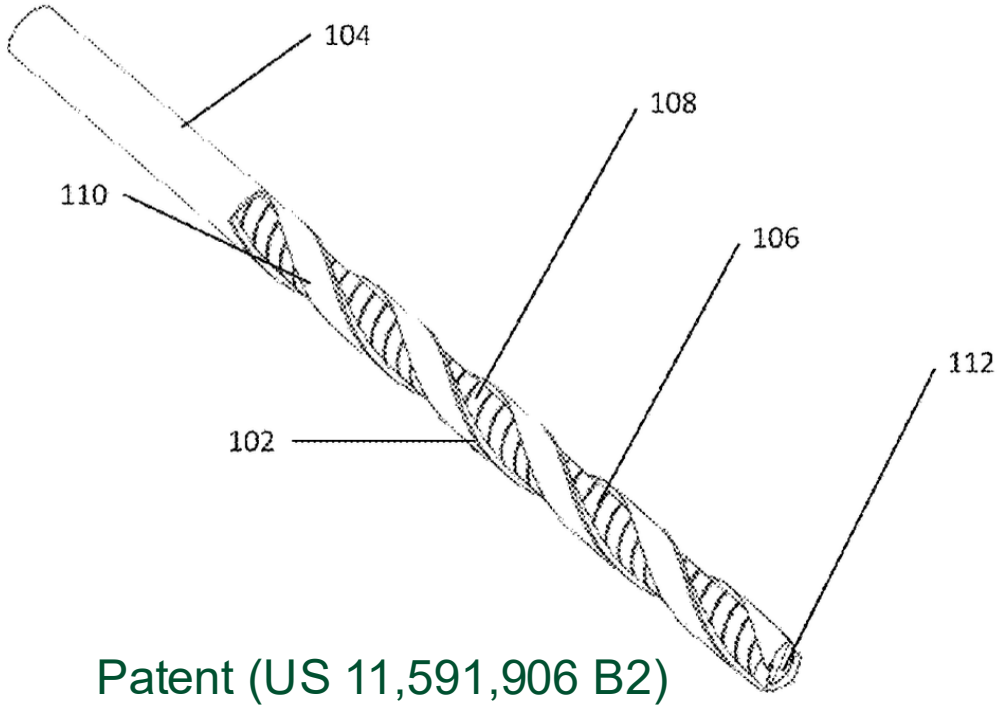
- Lacked vapor containment or flow management
- Knowledge gap in how vapor could flow through or around a drill bit.
- Further research into existing JPL and NASA drilling technologies was needed before continuing design work.





Research Discovery

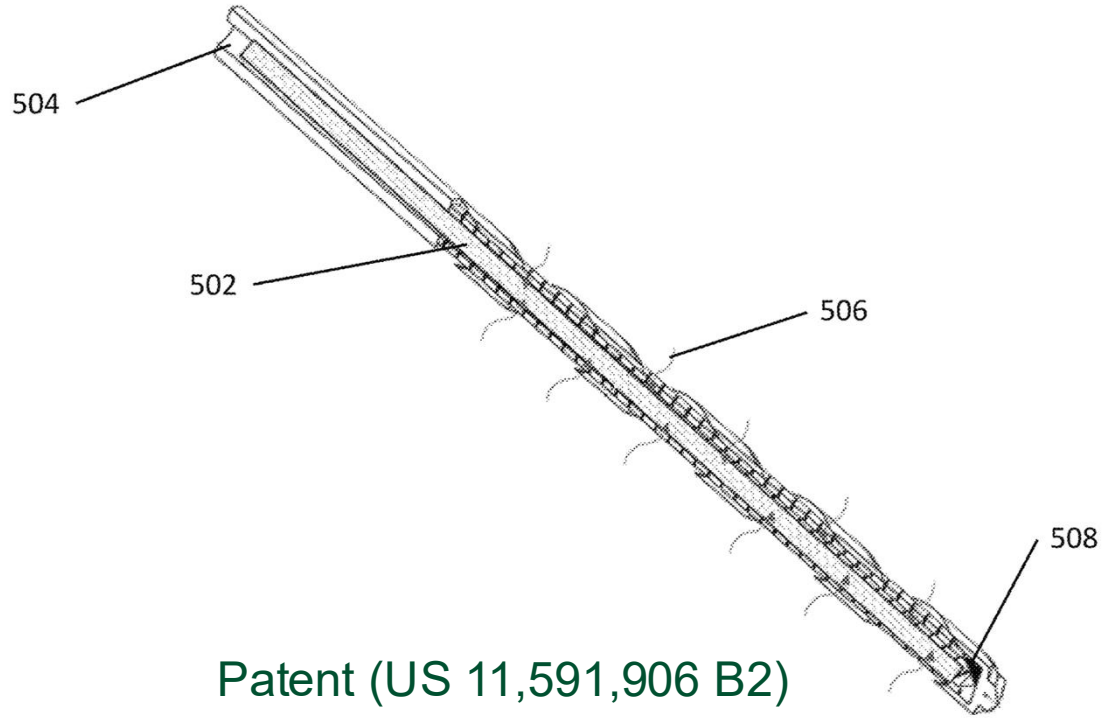
- Researched NASA’s drilling system patents, after initial tests
- Found JPL patent: 3D-printed drill with a porous core
- Aligned closely with goal of capturing and measuring vaporized water.



Patent (US 11,591,906 B2)

Design Pivot

- Set up meeting with the Patent Engineer
- Discussed patent concept to further understand
- Next iteration, integrates vapor channels



Patent (US 11,591,906 B2)

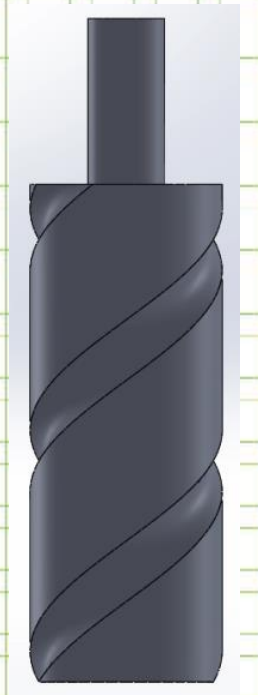
Design 2

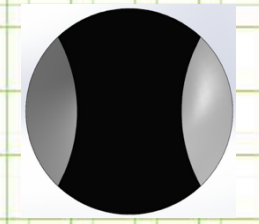
Porous Core Concept

- Porous core for vapor flow
- Cutting-tip angle reduced 75° → 40° for increased regolith engagement.
- Single helix maintained for material transport.

Testing Observations

- Vapor flow through core (concept validated)
- No heating → no vaporization
- Simulant escaped channels during retraction







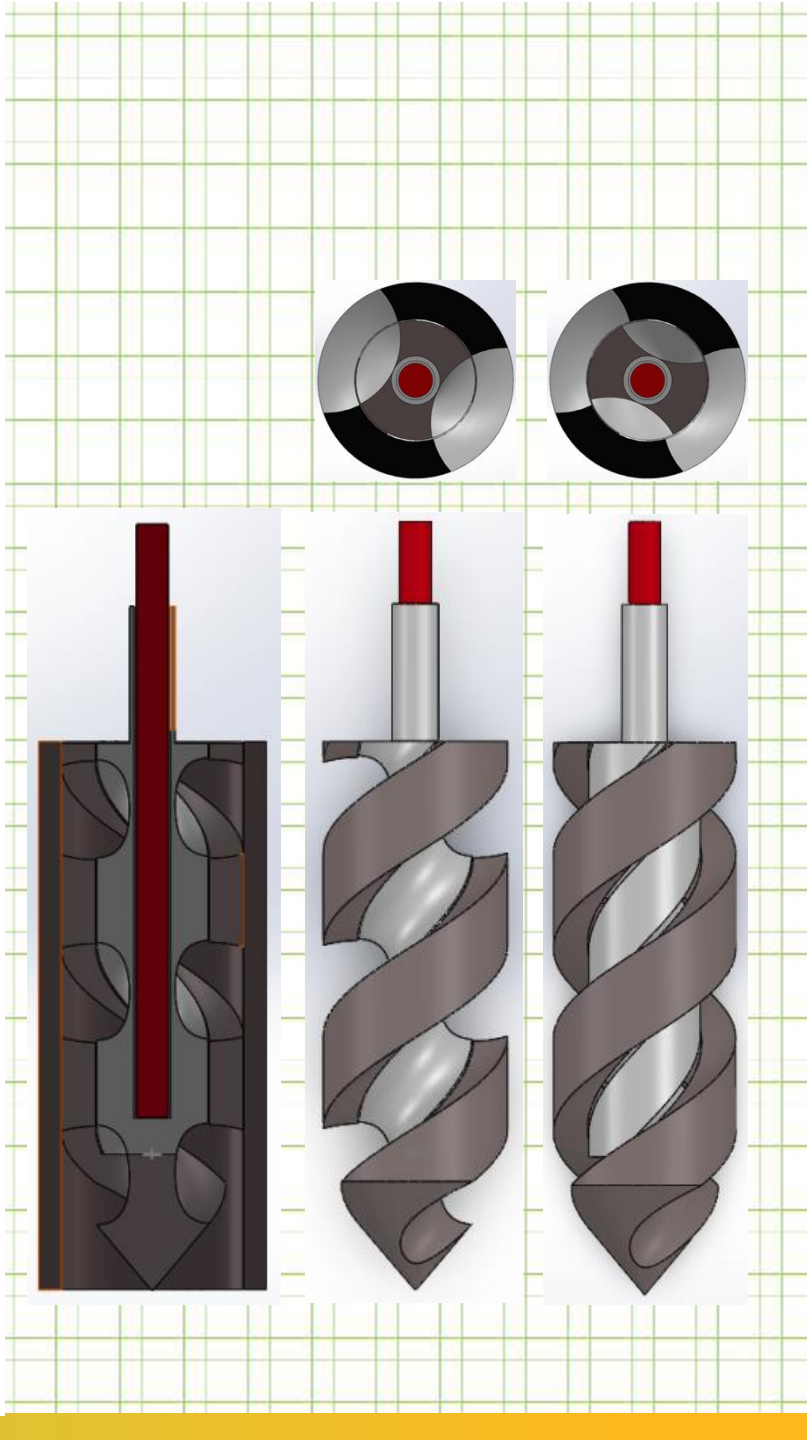
Design 3

Heated Retrieval Testing System

- Integrated central heating element within porous core
- After sample collection, drill retracts into enclosure sheath for vapor capture
- Combined porous (vapor flow) and solid (torque transfer) sections
- Tested potential for independent rotation between core and shell

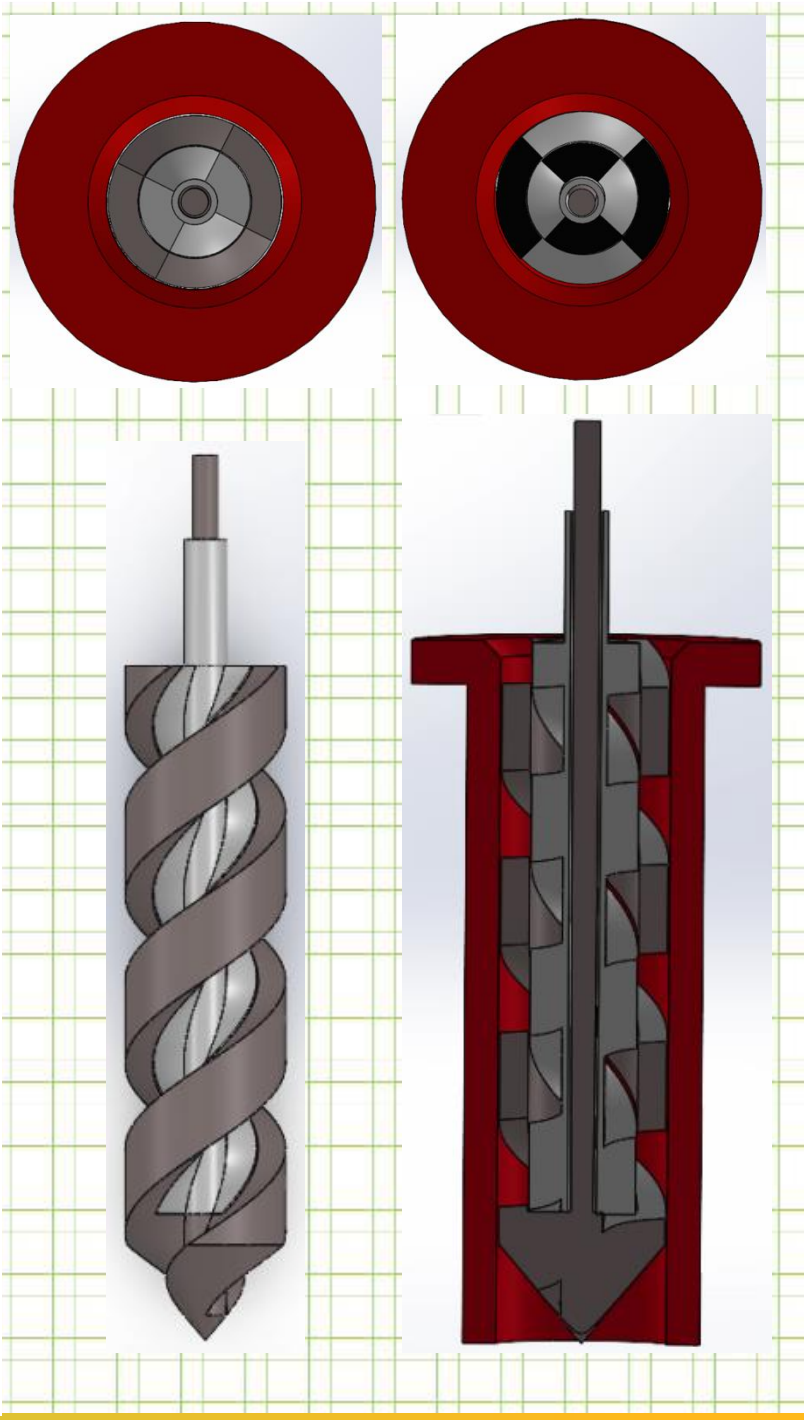
Testing Observations

- Vapor generation validated: heating can successfully vaporize water
- Offset geometry created a small enclosed vapor chamber during retraction
- Insight: independent rotation between solid & porous sections can enable controlled vapor chamber design



Design 4
Dual-Offset Enclosed Sheath System

- Dual-offset rotation (90° phase shift) → forms a sealed vapor chamber.
- Quarter-circle cross-section → smoother drilling, improved contact
- Auger extension to maintain a consistent sample volume
- Stationary heated sheath: Heats captured sample for vapor release during bit retraction



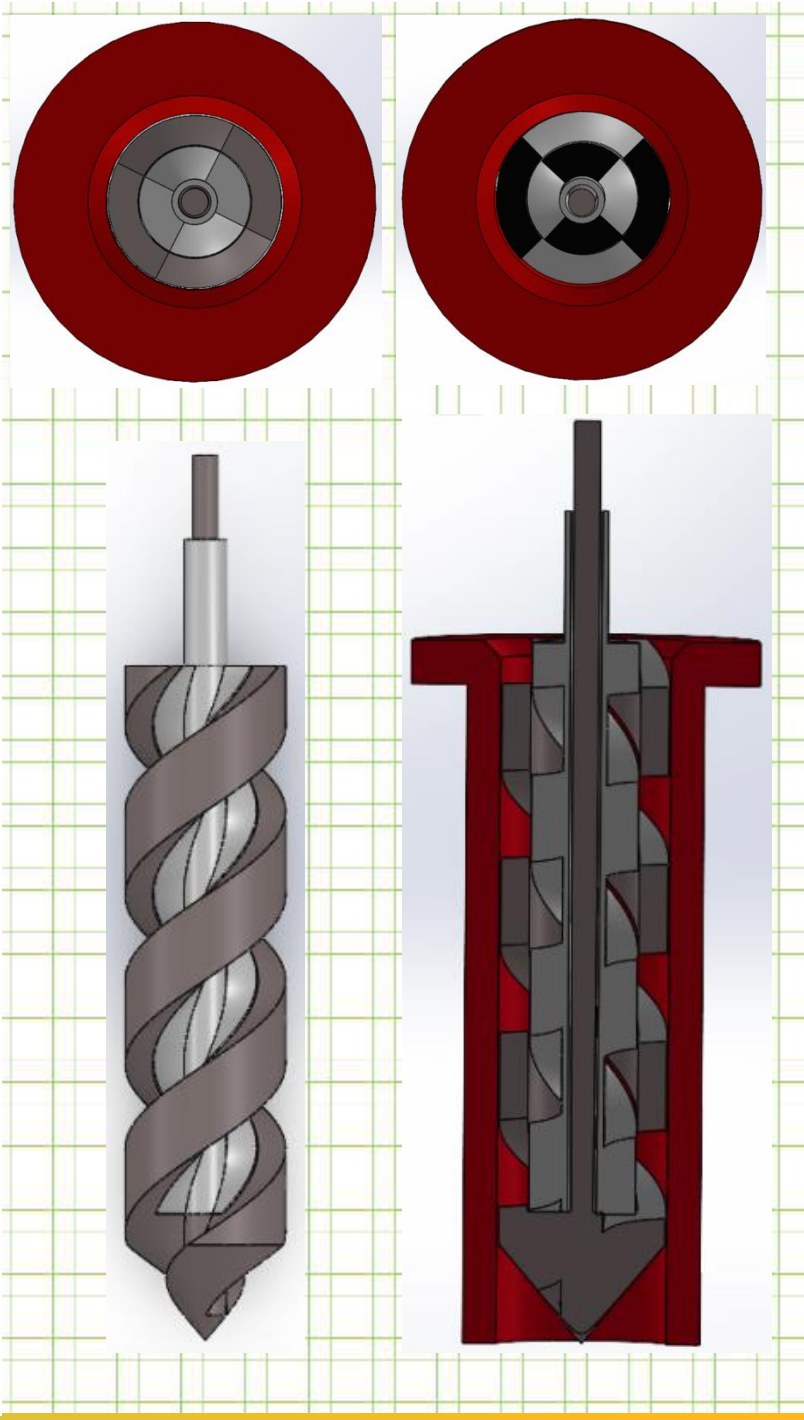
Design 4
Dual-Offset Enclosed Sheath System

Operation Sequence

1. Drill: Both sections rotate together to collect regolith.
2. Offset: Outer section rotates 90°, sealing the chamber.
3. Heat: Bit retracts into the heated sheath → vaporization.
4. Measure: Vapor exits through the porous core for isotopic analysis.

Key Outcome

- Validated dual-offset concept for mechanical vapor enclosure.
- Quarter circle cross-section improved consistent sample volume.



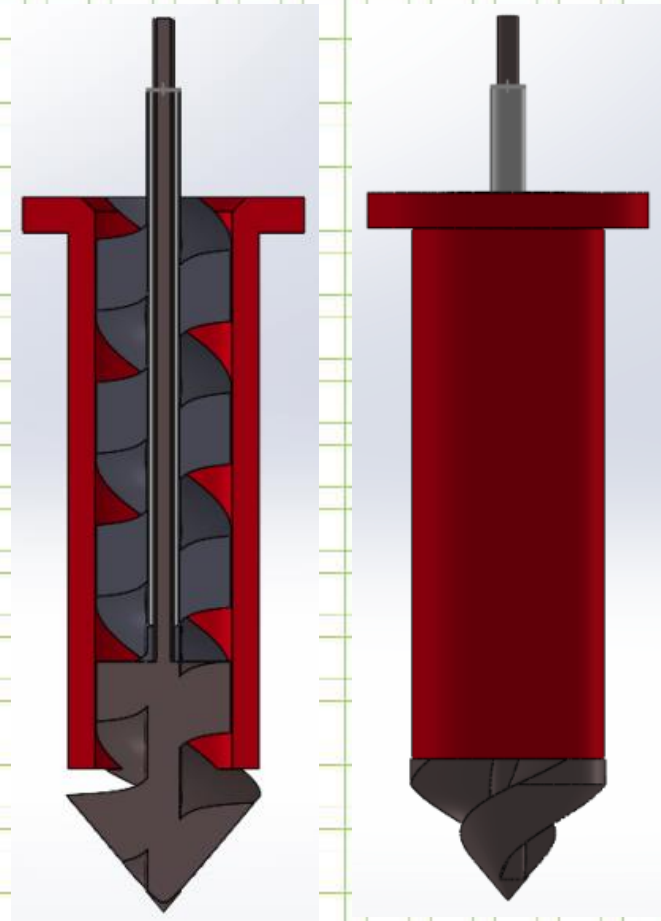
Design 5

Mobile Sheath and Optimized Porosity System

- Mobile heated sheath moves with drill bit → improving sample collection

- Optimized porosity:
  - Auger inner wall → Porous for vapor flow
  - Drill Bit Tip Axel → Solid for strength and torque transfer

- Enlarged the drill bit tip to create clearance for the sheath and maintain smooth entry into the regolith.



Design 5

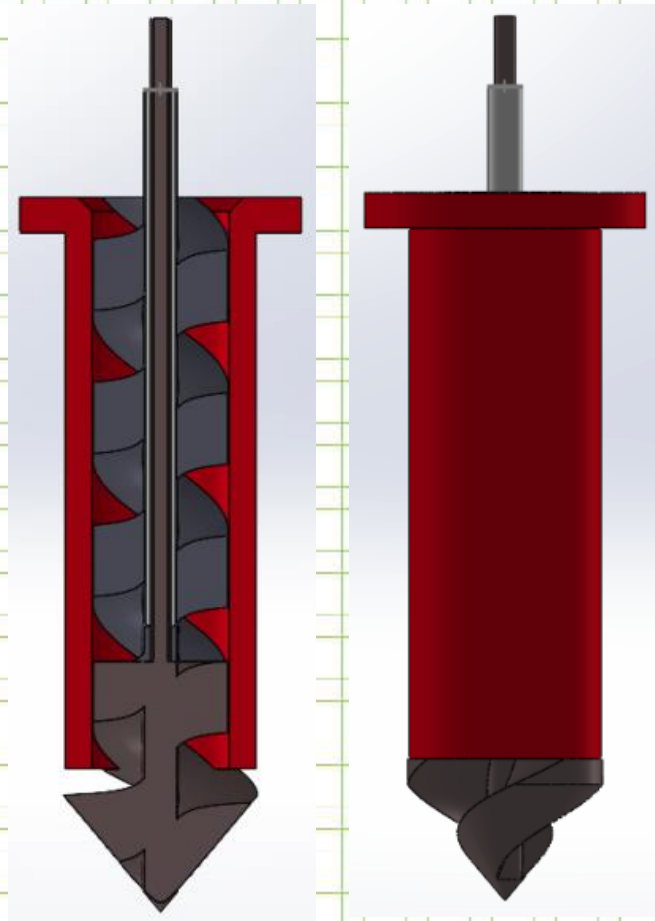
Mobile Sheath and Optimized Porosity System

Operation Sequence

1. Drill: Both sections rotate together to collect regolith.
2. Offset: Auger & tip offset 90°, sealing the chamber.
3. Heat → vaporization.
4. Measure: Vapor exits through the porous core for isotopic analysis.

Testing Observations

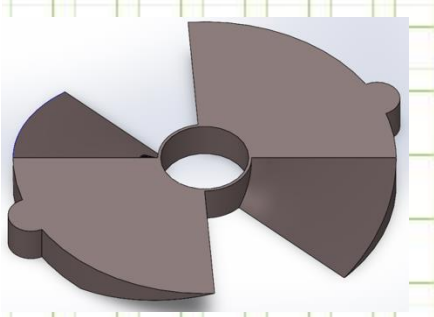
- Improved heat distribution and uniform vapor release
- Moving sheath reduced sample loss

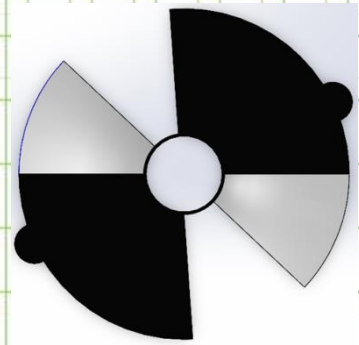


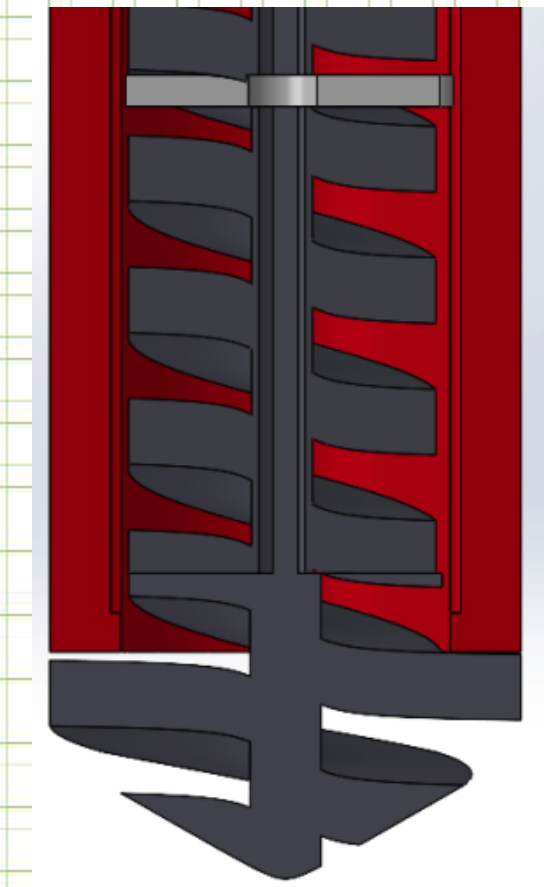
Design 6

Integrated Sample Ejection Mechanism

- Added passive ejection plate aligned with helical sheath tracks.
- Plate shares the drill's cross-section geometry, translating vertically during rotation.
- Integrated into mobile heated sheath from Design 5 for thermal vapor extraction







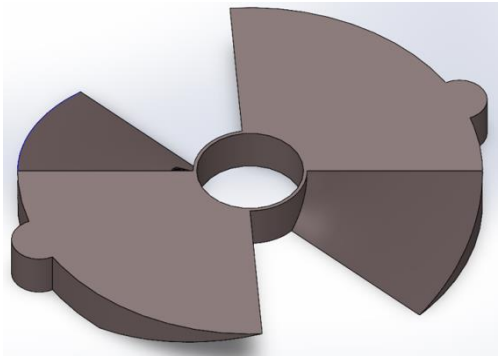
Operation Sequence

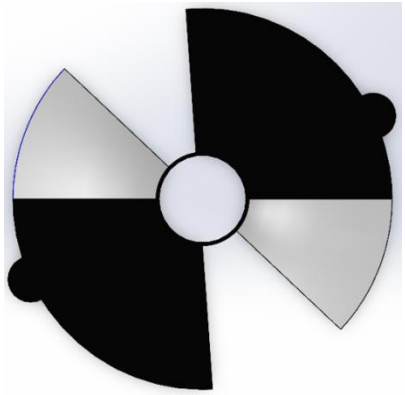
1. Drill and sheath collect regolith.
2. Drill bit tip and auger offset 90* → seal chamber
3. Heat sheath → vaporize any water molecules
4. Vapor exits through the porous core for isotopic analysis
5. Ejector plate descends → clears chamber for next test

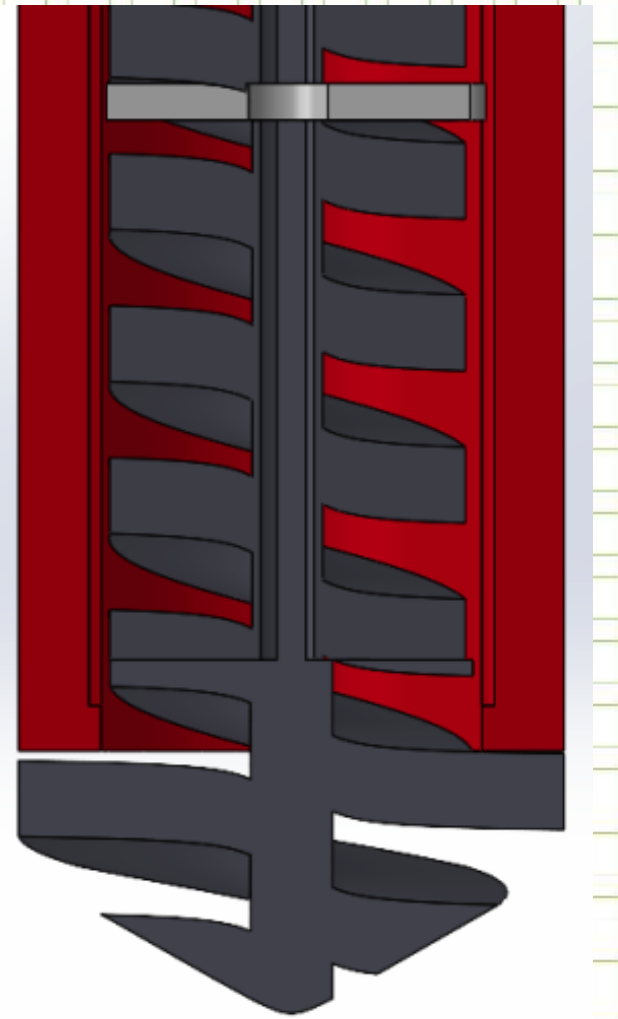
Design 6
Integrated Sample Ejection Mechanism

Key Outcomes

- Automated ejection using existing rotation
- Enables repeatable testing cycles
- Maintained mechanical simplicity and vapor performance





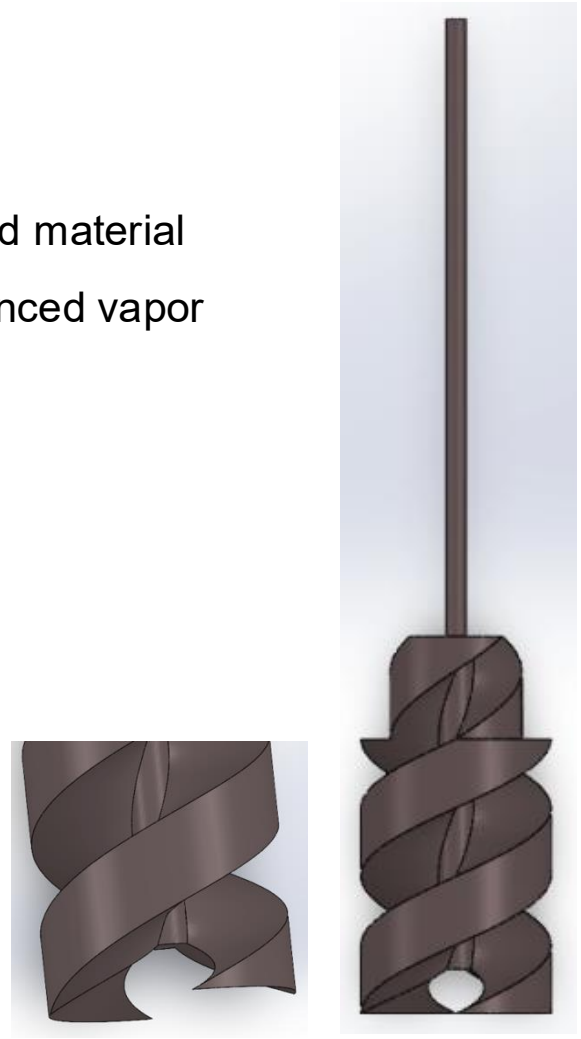


Design 7
Drill Bit Tip Optimization

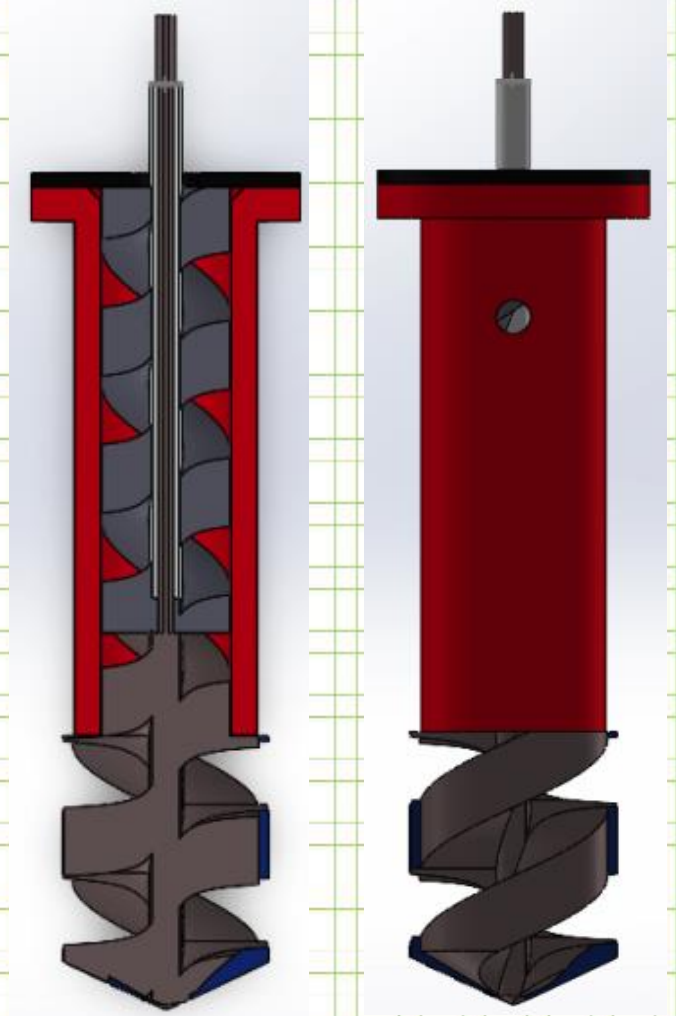
- Refine drill bit tip geometry for improved material penetration, reduced torque, and enhanced vapor exposure during in-sheath heating.

Key Outcomes

- Core-saw cutting edge → smoother regolith entry.
- Pulverizing edge cutter → increases fine particle generation.
- Central bore maintains → preserves vapor flow and sample clearance.



Coring Saw concept



Pulverizing edge cutter added

Future Work and Next Steps

Short Term Goals (0-6 Months)

- Integrate heating element for thermal vapor extraction testing.
- 3D print metal prototype and perform full-scale tests at JPL in a vacuum-lunar environment.
- Conduct extended lab testing with regolith simulant to verify in-situ performance.

Long Term Goals (6-12+ Months)

- Validate performance and vapor capture efficiency under lunar-like conditions.
- Refine drill geometry and porosity to improve vapor extraction and durability.
- Develop a fully automated cycle: drill → heat → detect → eject → reset.
- Integrate with ISRU systems for water and hydrogen isotopic quantification.



Thank You!